

Plant Kingdom

SHORT NOTES

Classification Systems

Artificial Classification System:

- ❖ Given by Carolus Linnaeus.
- ❖ Based mainly on vegetative characters or on the androecium structure.
- ❖ Separated the closely related species since they were based on a few characteristics.
- ❖ Gave equal weightage to vegetative and sexual characteristics.

Natural Classification System:

- ❖ Given by George Bentham and Joseph Dalton Hooker.
- ❖ Based on natural affinities among the organisms.
- ❖ Consider both external and internal features.

Phylogenetic Classification System:

- ❖ Based on evolutionary relationships between the various organisms.
- ❖ Assumes that organisms belonging to the same taxa have a common ancestor.

Numerical Taxonomy

- ❖ Carried out using computers and based on all observable characteristics.
- ❖ Each character is given equal importance and at the same time hundreds of characters can be considered.

Cytotaxonomy

- ❖ Based on cytological information like chromosome number, structure, behaviour.

Chemotaxonomy

- ❖ Uses the chemical constituents of the plant to resolve confusions.

Algae

Classification of Algae			
Features	Chlorophyceae	Phaeophyceae	Rhodophyceae
Common name	Green algae	Brown algae	Red algae
Pigment	Chl <i>a, b</i>	Chl <i>a, c</i> and fucoxanthin	Chl <i>a, d</i> and <i>r</i> -phycoerythrin
Reserve food	Starch	Laminarin, Mannitol	Floridean starch
Flagella	2-8, equal, apical	2, unequal, lateral	Absent
Examples	<i>Chlamydomonas</i> , <i>Volvox</i> etc.	<i>Ectocarpus</i> , <i>Sargassum</i> etc.	<i>Polysiphonia</i> , <i>Gracilaria</i> etc.

Bryophytes

- ❖ Commonly called as **amphibians** of plant kingdom.
- ❖ Includes Liverworts and mosses.
- ❖ Vascular tissues are **absent**.
- ❖ Lack true stem, leaves and roots.
- ❖ Main plant body is **gametophyte**. **Sporophyte** is dependent on gametophyte for nutrition.

- ❖ Plant body attached to substratum by unicellular or multicellular **rhizoids**.
- ❖ Water is essential for fertilization.
- ❖ Sex organs are **multicellular**. Male-**antheridium** and female-**archegonium**.
- ❖ Fertilization produces zygote which develops into multicellular sporophyte.

Liverworts

- ❖ Grow usually in moist and shady habitats.
- ❖ Plant body is thalloid which is dorsiventral and closely appressed to the substrate.
- ❖ Leafy members have tiny leaf-like appendages in two rows on the stem-like structures.
- ❖ Asexual reproduction through fragmentation.
- ❖ **Gemmae** are green, multicellular asexual buds.
- ❖ Sporophyte is formed as a result of sexual reproduction.
- ❖ Sporophyte is differentiated into foot, seta and capsule.
- ❖ After meiosis, spores are produced within the capsule which germinate to form free-living gametophytes. **E.g.,** *Marchantia*

Mosses

- ❖ Protonema is the first stage of gametophyte that develops directly from spore.
- ❖ Leafy stage is second stage develops from secondary protonema as lateral bud.
- ❖ Vegetative reproduction through fragmentation and budding in secondary protonema.
- ❖ Sexual reproduction present.
- ❖ Have an elaborate mechanism of spore dispersal. **E.g.,** *Funaria*, *Polytrichum* and *Sphagnum*.

Pteridophytes

- ❖ Includes horsetails and ferns.
- ❖ First vascular plants without seeds.
- ❖ Main plant body is sporophyte and distinguishes into root, stem and leaves.
- ❖ Reproduction is of vegetative, asexual and sexual type.
- ❖ Male gametes are flagellated.
- ❖ Sporangia produce spores by meiosis in spore mother cells.
- ❖ Spores germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called **prothallus**.
- ❖ Requires water for fertilisation.
- ❖ Development of the zygotes into young embryos take place within the female gametophytes which represents a precursor to the seed habit. **E.g.,** *Psilotum*, *Lycopodium*, *Selaginella* etc.

Gymnosperms

- ❖ Vascular plants with **naked seeds**.
- ❖ Redwood tree *Sequoia* is one of the tallest tree species.

- ❖ In *Pinus* roots have fungal association in the form of *mycorrhiza*.
- ❖ Coralloid roots are found in *Cycas* which are associated with N_2 -fixing cyanobacteria.
- ❖ Male and female cones may be borne on same tree as in *Pinus* or in different tree as in *Cycas*.
- ❖ Male and the female gametophytes do not have an independent free-living existence.
- ❖ Zygote is formed after fertilisation of pollen grain and ovule.
- ❖ Zygote develops into an embryo and the ovules into seeds. These seeds are naked.

Angiosperms

- ❖ Angiosperms are flowering plants and have seeds enclosed in fruits and have well-developed flowers containing reproductive organs.
- ❖ The size of angiosperms vary from as small as *Wolffia* with 0.1 cm to tall trees of *Eucalyptus*, which may grow as long as 100 m.
- ❖ Angiosperms are divided into two classes; Monocotyledons, Dicotyledons
- ❖ The flower contains 4 parts; calyx (sepals), corolla (petals), androecium and gynoecium (male and female reproductive parts respectively).
- ❖ An ovule bears embryo sac.
- ❖ Double fertilization is the characteristic of angiosperms.
- ❖ Two male gametes enter the embryo sac through the pollen tube. The egg and one male gamete fuse together to form a diploid zygote, the process is known as syngamy.
- ❖ The second male gamete fuses with the diploid secondary nucleus, the process is known as triple fusion and triploid endosperm is formed. It is known as double fertilization because two fusions happen, i.e. syngamy and triple fusion.
- ❖ Ovaries develop into fruit and ovules develop into a seed.

